[CORMEN] Intro à l'algo

27/09/2023
SDA
Rappels

Pb 1.1.

Chap. 3 (Croissance des fonctions)
Pb 3.1., 3.2., 3.3.

Chap 4 (Récurrences)

Annexes A (sommations)
C (Dénombrement et proba)

Fonctions classiques

☆ Parties entières (ceil et floor)

$\forall x \in \mathbb{R}$,

$$x-1 < \lfloor x \rfloor \leq x \leq \lceil x \rceil < x+1$$

$$\left(\left\lfloor \frac{n}{2} \right\rfloor + \left\lceil \frac{n}{2} \right\rceil = n\right)$$

☆ Logarithmes

notation : $\lg n := \log_2 n = \frac{\ln n}{\ln 2}$

(logarithme binaire)

$\lg 2^k = k$

$\forall a,b \in \mathbb{R}_+^*, \forall n \in \mathbb{Z}$, $\lg(a \times b) = \lg a + \lg b$

$\lg\left(\frac{1}{a}\right) = -\lg a$

$\lg a^n = n \lg a$

☆ "exponentielles"

$\forall a \in \mathbb{R}^*, \forall n,m \in \mathbb{Z}$, $a^0 = 1$ $\quad (a^m)^n = a^{m \cdot n} = (a^n)^m$

$a^1 = a$ $\quad a^m \cdot a^n = a^{m+n}$

$a^{-1} = \frac{1}{a}$

☆ Factorielle

$$n! = n \times (n-1)! = \prod_{i=1}^{n} i$$

Formule de STIRLING

$$n! = \sqrt{2\pi n}\left(\frac{n}{e}\right)^n \left(1 + \Theta\left(\frac{1}{n}\right)\right)$$

en particulier : $n! = o(n^n)$

$n! = \omega(2^n)$

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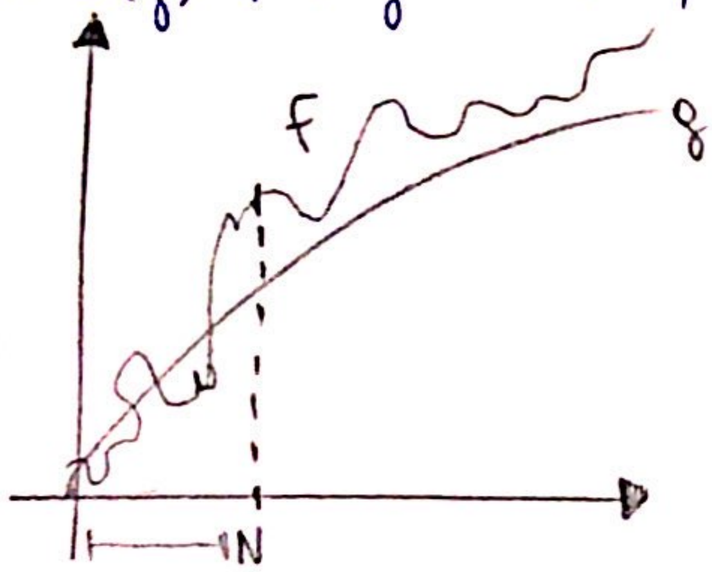
☆ Notations asymptotiques

Soient $f, g : \mathbb{N} \to \mathbb{N}$

$F = O(g)$ lorsque $\exists c \in \mathbb{R}_+^*, \exists N \in \mathbb{N}, \forall n \geq N \quad F(n) \leq c \cdot g(n)$

$F = \Omega(g)$ lorsque $\quad c \cdot g(n) \leq F(n)$

(ii) $F = O(g) \iff g = \Omega(F)$



$f = \Omega(g)$

$F = O(g)$ et $F = \Omega(g) \iff F = \Theta(g)$

$F = O(g)$ lorsque $\lim_{n \to +\infty} \frac{F(n)}{g(n)} = 0$ dans ce cas là, $\iff g = \omega(F)$

☆ $\lg^* n := \min \{ i \in \mathbb{N} : \lg^{(i)} n \leq 1 \}$

$\lg^* 2 = 1$

$\lg^* 4 = 2$

$\lg^* 16 = 3$

$\lg^* 2^{65536} = 5$

$\lg^*(2^{2^2}) = \lg^3(65536) = 4$

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